Alumne Daniel Lopez Martí

P2P “geneclimich”

NIVELL 1

Exercici 1

Creamos la base de datos donde se van a guardar las tablas

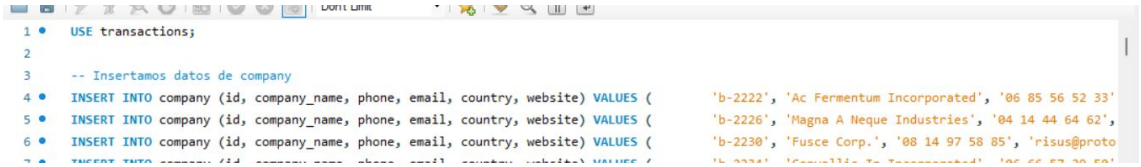
```sql
-- Creamos la base de datos
CREATE DATABASE IF NOT EXISTS transactions;
USE transactions;
```

Creamos las tablas vacías:

```sql
-- Creamos la tabla company
CREATE TABLE IF NOT EXISTS company (
    id VARCHAR(15) PRIMARY KEY,
    company_name VARCHAR(255),
    phone VARCHAR(15),
    email VARCHAR(100),
    country VARCHAR(100),
    website VARCHAR(255)
);
```

```sql
-- Creamos la tabla transaction
CREATE TABLE IF NOT EXISTS transaction (
    id VARCHAR(255) PRIMARY KEY,
    credit_card_id VARCHAR(15) REFERENCES credit_card(id),
    company_id VARCHAR(20),
    user_id INT REFERENCES user(id),
    lat FLOAT,
    longitude FLOAT,
    timestamp TIMESTAMP,
    amount DECIMAL(10, 2),
    declined BOOLEAN,
    FOREIGN KEY (company_id) REFERENCES company(id)
);
```

Un cop creades insertem les dades



```
USE transactions;

-- Insertamos datos de company
INSERT INTO company (id, company_name, phone, email, country, website) VALUES (       'b-2222', 'Ac Fermentum Incorporated', '06 85 56 52 33'
INSERT INTO company (id, company_name, phone, email, country, website) VALUES (       'b-2226', 'Magna A Neque Industries', '04 14 44 64 62',
INSERT INTO company (id, company_name, phone, email, country, website) VALUES (       'b-2230', 'Fusce Corp.', '08 14 97 58 85', 'risus@proto
```

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.

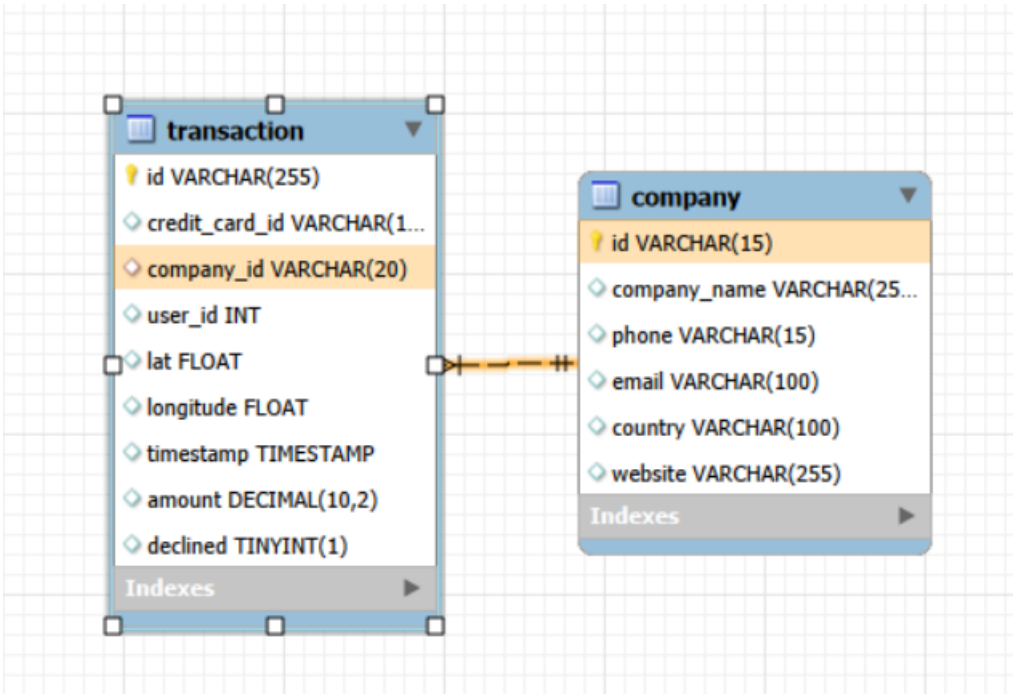


Tabla Transaction:

-Esta tabla registra las transacciones financieras realizadas por los usuarios entre los años 2015 y 2025.

COLUMNAS:

id VARCHAR(225): Clave Primaria, sin conectar a ninguna, Identificador único para cada registro de transacción. Texto, con un máximo de 225 caracteres.

credit_card_id VARCHAR(15): Identificador de la tarjeta de crédito utilizada. Texto con un máximo de 15 caracteres Texto con máxima longitud de 15 caracteres.

company_id INT: Clave Foránea. Conecta directamente con la **tabla Company vinculada a 100 empresas distintas** Numero entero de longitud indefinida.

user_id INT: Identificador único en número entero del usuario que realizó la operación.Numero entero de longitud indefinido.

lat DECIMAL: Latitud geográfica en números decimales donde se efectuó la transacción.

longitude DECIMAL: Longitud geográfica en números decimales donde se efectuó la transacción.

timestamp(TIMESTAMP): Hora de la transacción, solo acepta el formato único para mostrar momento precioso de la misma.

amount DECIMAL(12, 2): Monto de la transacción con dos decimales y un valor máximo de 9999999999,99.

declined TINYINT: Estado de la transacción optimizado en espacio que solo acepta los valores enteros 1 o 2 para actuar como un sí o un no.

Tabla Company:

Esta tabla almacena los datos maestros de las 100 empresas relacionadas con las operaciones.

company_id INT: Clave Primaria. Identificador único de la empresa que conecta con las transacciones.

Phone contactVARCHAR(15):teléfono de contacto,es un texto de una máxima extensión de 15 caracteres.

Email VARCHAR(100): direccion de correo electrónico decontacto con una extensión máxima de 100 caracteres.

countryVARCHAR(100) :país donde esta la empresa, cuenta como tenxto de un máximo de extensión de 100 caracteres.

WebsideVARCHAR(250):dominio Webb de la empresa, texto máximas extensión 255 caracteres

Exercici 2(Utilitzant JOIN realitzaràs les següents consultes:)

- 2.1Llistat dels països que estan generant vendes.

```
SELECT DISTINCT company.country
FROM company
JOIN transaction ON company.id=transaction.company_id
WHERE transaction.declined=0
and transaction.amount >=0
and company_name IS NOT NULL;
```

country
Netherlands
Sweden
Ireland
United States
Belgium

Result 8 | Read Only

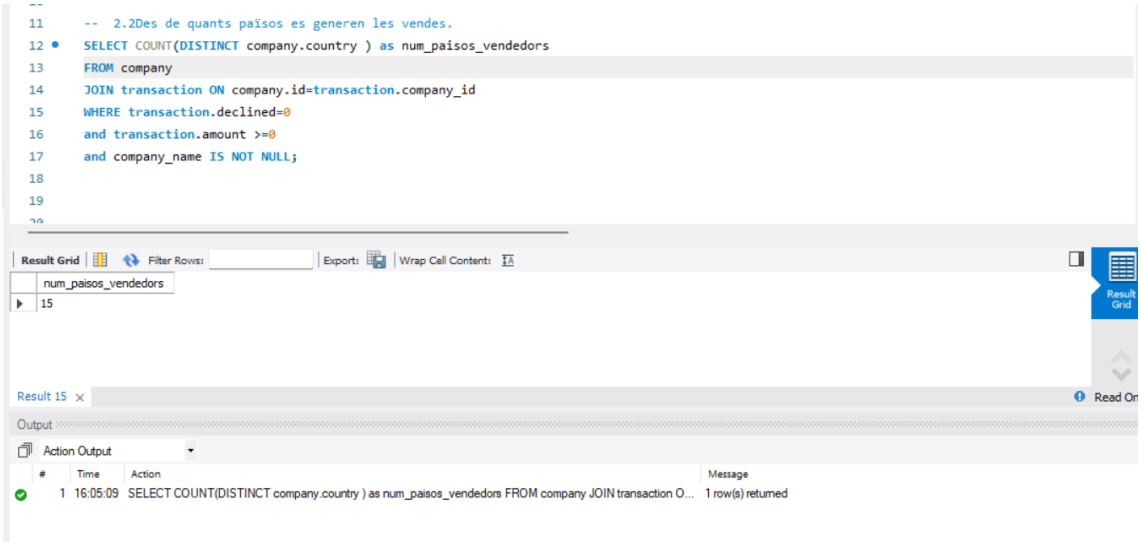
Output

Action Output

#	Time	Action	Message
1	15:55:44	SELECT DISTINCT company.country FROM company JOIN transaction ON company.id=transaction.company_i...	15 row(s) returned

En l'activitat2.1 em de tenir en compte que el amount pot ser de0 degut a un descompte del 100% i el declined ha de estar a 0 per mostra que s'ha efectuat la venta realmente, tambe incluim un IS NOT NULL per eliminar castellas buidas que suman al resultat

- 2.2Des de quants països es generen les vendes.



Em de modificar lleugerament el exercici anterior i tenim el resultat

- 2.3Identifica la companyia amb la mitjana més gran de vendes.

```sql
SELECT company.company_name
FROM company
JOIN transaction ON company.id = transaction.company_id
WHERE transaction.declined = 0
  AND transaction.amount >= 0
GROUP BY company.company_name
ORDER BY AVG(transaction.amount) DESC
LIMIT 1;
```

company_name
Ac Fermentum Incorporated

Result 24

Output

#	Time	Action	Message
1	16:14:21	SELECT company.company_name FROM company JOIN transaction ON company.id = transaction.company_id...	1 row(s) returned

Pose el AVG en el order BY per que nomes em de mostra el nom de la compañía

Exercici 3

Utilitzant només subconsultes (sense utilitzar JOIN):

- Mostra totes les transaccions realitzades per empreses d'Alemanya.

```sql
-- Mostra totes les transaccions realitzades per empreses d'Alemanya.
SELECT*
FROM transaction
WHERE amount>=0
    and declined=0
    and company_id in(
            SELECT id
```

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
00138D3B-206D-4C03-94B7-63A2676EB9B4	CcS-4899	b-2222	318	41.3781	12.447	2020-03-25 10:43:43	426.36	0
0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9	CcS-5070	b-2222	489	41.3814	2.18176	2020-12-17 18:15:37	316.90	0
00201A11-2E62-44C4-941D-198FC8DB77F0	CcU-3512	b-2222	193	55.5704	-3.65129	2021-01-22 23:44:27	453.04	0
00235618-0A5C-4D49-9DCB-B3A9405D8923	CcS-8137	b-2222	3556	59.8421	18.729	2020-09-09 15:43:19	263.14	0
005A5A7B-1F1A-4B6C-9B15-1625A78C9C38	CcS-8998	b-2222	4417	41.1591	-8.63905	2024-05-15 09:10:11	442.01	0

transaction 29

Output

#	Time	Action	Message
1	16:27:18	SELECT* FROM transaction WHERE amount>=0 and declined=0 and company_id in(SELECT id FR...	13269 row(s) returned

- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```
41 •  SELECT company_name
42    FROM company
43    WHERE id IN (
44        SELECT company_id
45        FROM transaction
46        WHERE declined =0
47        GROUP BY company_id
48        HAVING AVG(amount)>(
49            SELECT AVG(amount)
50            FROM transaction));
51
52
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name
Ac Fermentum Incorporated
Magna A Neque Industries
Fusce Corp.
Ante Iaculis Nec Foundation
Donec Ltd
Nascetur Ridiculus Mus Inc.

company 3 ×

Output

Action Output

#	Time	Action	Message
1	18:42:29	SELECT company_name FROM company WHERE id IN (SELECT company_id FROM transaction WH...	48 row(s) returned

- Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

```
52 •  SELECT company_name
53    FROM company
54    WHERE NOT EXISTS (
55    SELECT company_id
56    FROM transaction);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name

company 5 ×

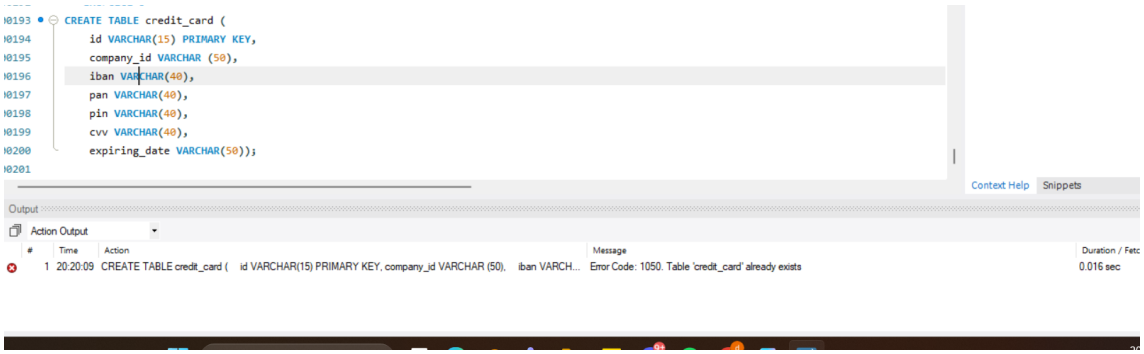
Output

Action Output

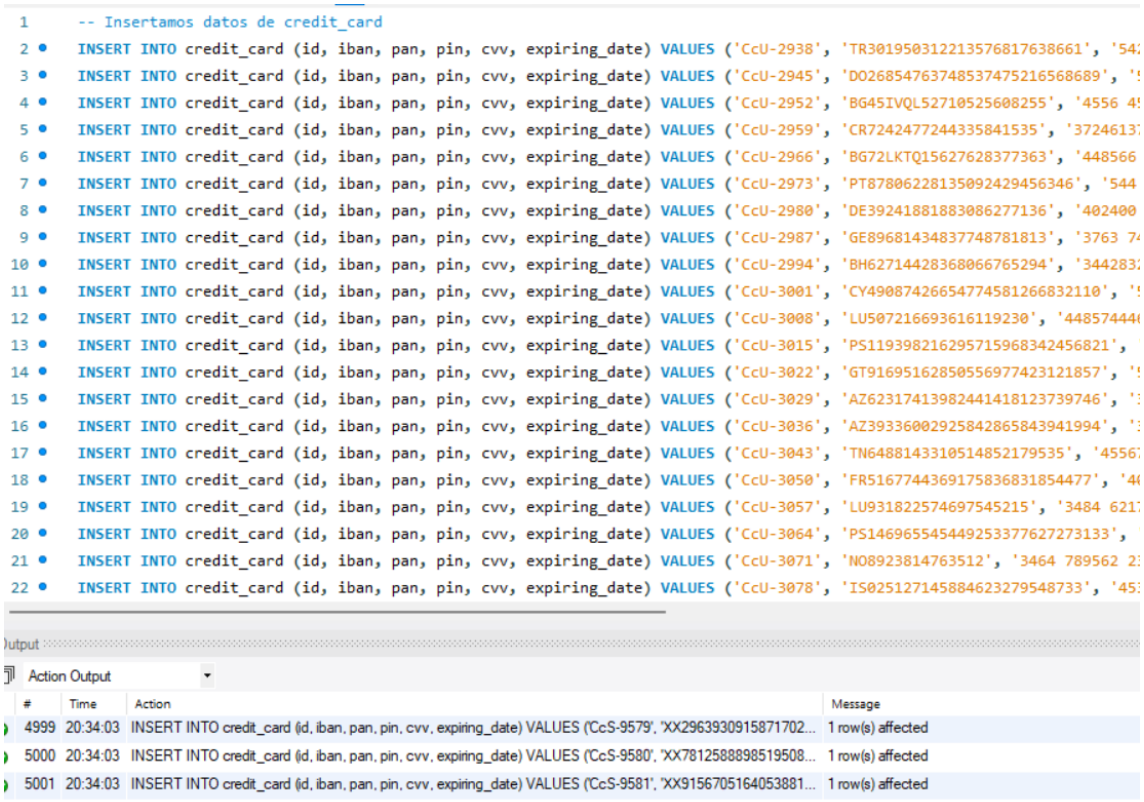
#	Time	Action	Message
1	18:45:49	SELECT company_name FROM company WHERE NOT EXISTS (SELECT company_id FROM transaction)	0 row(s) returned

Exercici 4

- La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.



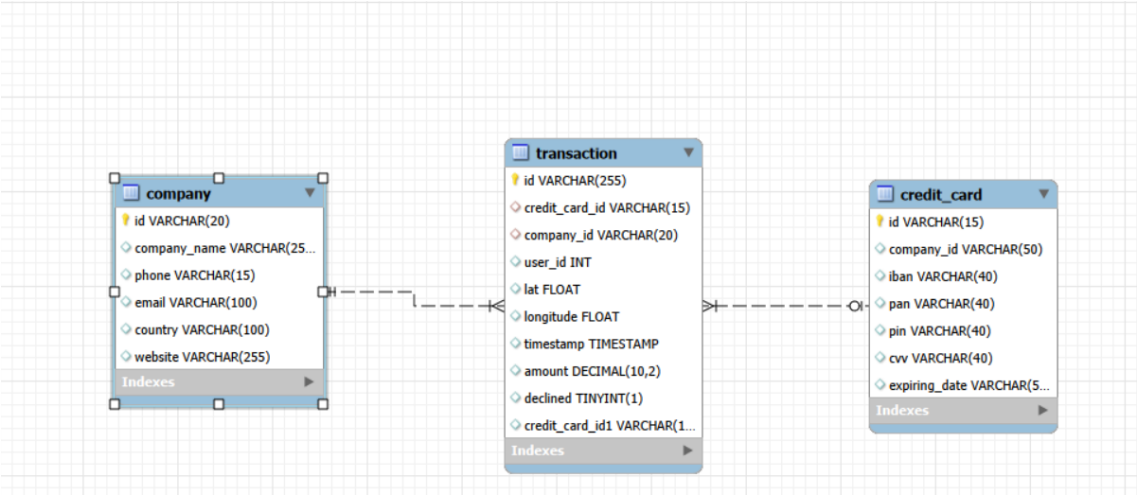
Creem la taula requerida(el error 1050 confirma que ja esta creada correctament)




```
    expiring_date VARCHAR(50),
    FOREIGN KEY (company_id) REFERENCES company(id)
);
UPDATE credit_card
JOIN transaction ON credit_card.id = transaction.credit_card_id
SET credit_card.company_id = transaction.company_id;
```

insertem les dades de la taula per la via de carregadel documenti donali el valor

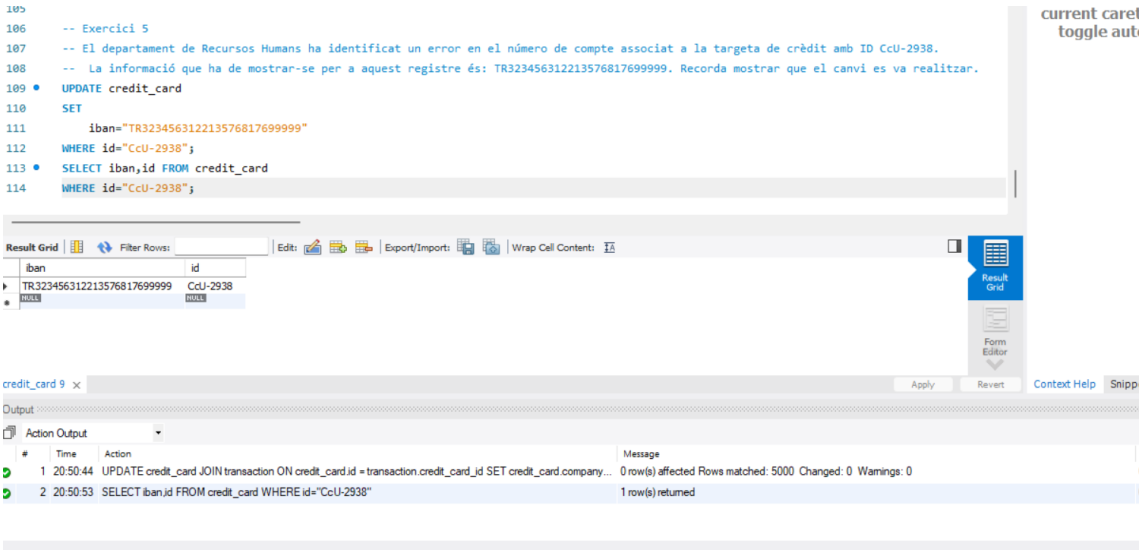
del forekey a través de la columna



Conectem les 3 taules correctamente, imodifiquem inclus la longitud de la primary key de company per que coincideixi pasan de (15) a (20)

Exercici 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.



Exercici 6

En la taula "transaction" ingressa una nova transacció amb la següent informació:

```
Id

108B1D1D-5B23-A76C-55EF-C568E49A99DD

credit_card_id

CcU-9999

company_id

b-9999

user_id

9999

lat

829.999

longitude

-117.999

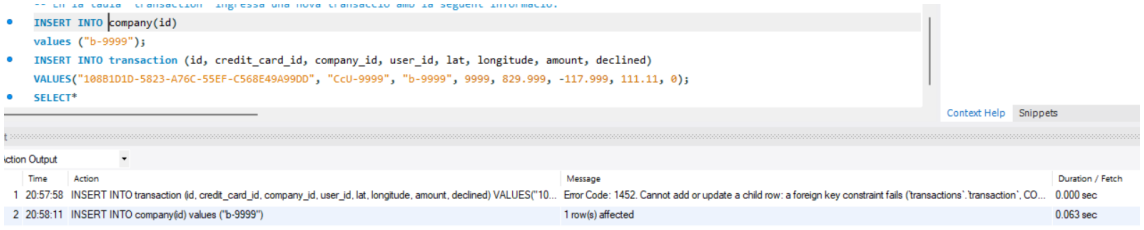
amount

111.11

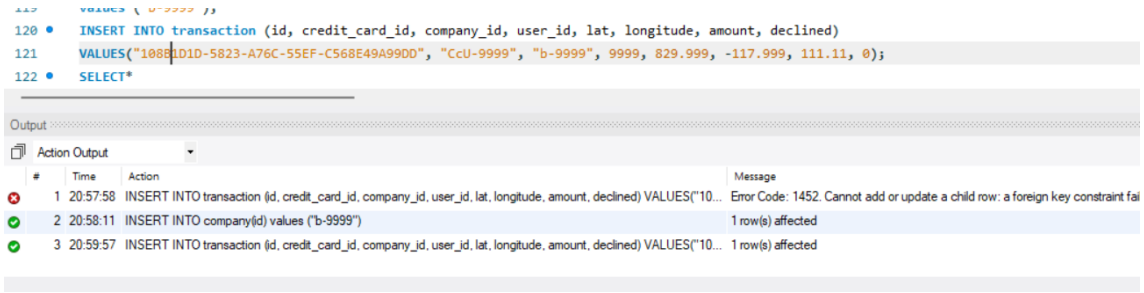
declined

0
```

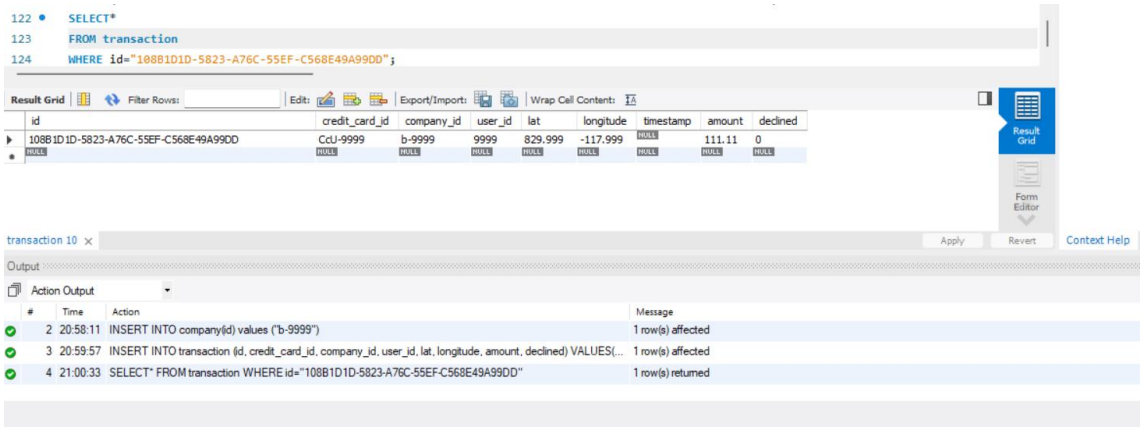
Primer insertem la company id per que es una foreign key que s'hauria de donar desde una altre taula



Afegim tot



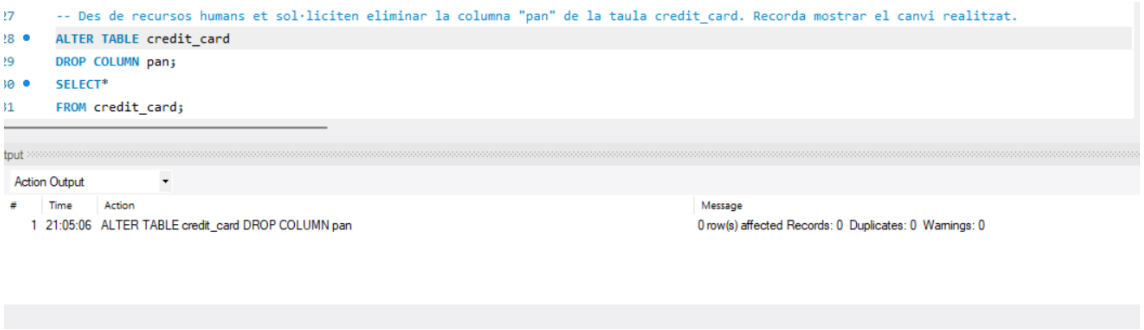
Comprobem el resultat



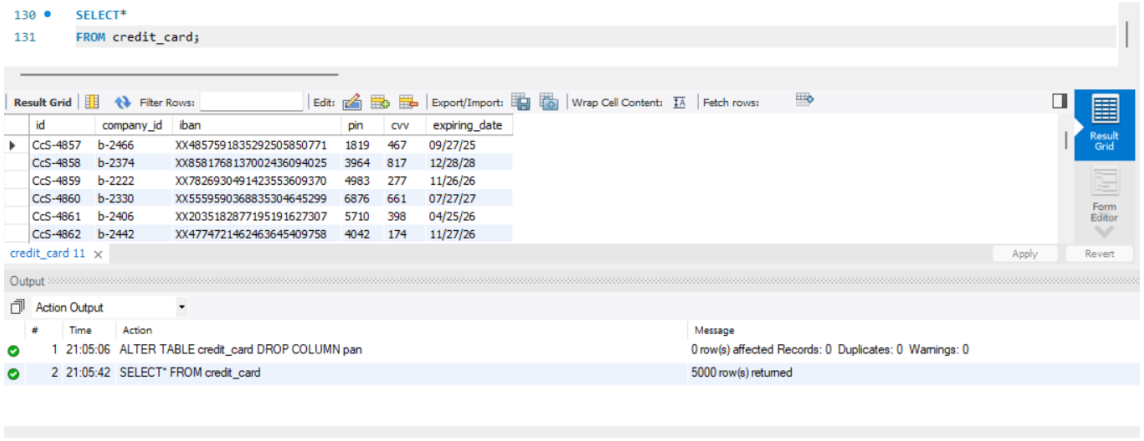
Exercici 7

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recorda mostrar el canvi realitzat.

Eliminem la columna



Comprobem resultat



EXERCICI 8

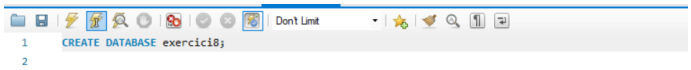
Descarrega els arxius CSV que **trobaràs a l'apartat de recursos**:

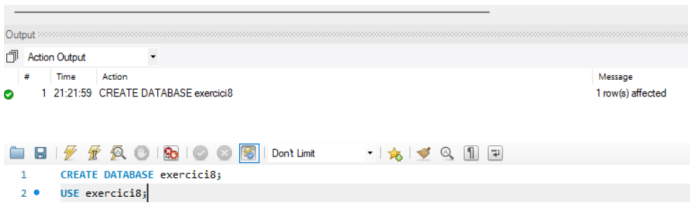
- american_users.csv

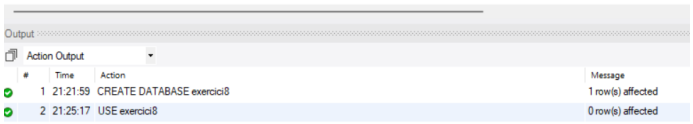
- european_users.csv
- companies.csv
- credit_cards.csv
- transactions.csv

Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:
La taula de products.csv l'utilitzarem més endavant.

Creem una base de dades buida i la usem







Creen les taules

```sql
CREATE TABLE transactions(
    id VARCHAR(250),
    card_id VARCHAR (20),
    business_id VARCHAR(250),
    timestamp VARCHAR(50),
    amount FLOAT,
    declined TINYINT,
    product_ids VARCHAR(200),
    user_id VARCHAR(255),
    lat FLOAT,
    longitude FLOAT,
    discount_amount FLOAT,
    tax_amount FLOAT,
    shipping_amount FLOAT,
    channel VARCHAR(30),
    campaign_id VARCHAR(30),
    device_type VARCHAR (50),
    is_international TINYINT,
    decline_reason VARCHAR (50),
    distance_km FLOAT);
```

Output

Time	Action	Message
21:28:13	CREATE TABLE transactions(id VARCHAR(250), card_id VARCHAR (20), business_id VARCHAR(250), timesta...	0 row(s) affected

Carregem les dades desde una carpeta que el programi acepti sense alterar la seguretat del programa

```sql
    SET FOREIGN_KEY_CHECKS = 0;
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__transactions.csv'
INTO TABLE transactions
FIELDS TERMINATED BY ';'
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES
(id, card_id, business_id, timestamp, amount, declined, product_ids, user_id, lat, longitude, discount_amount, tax_amount, shipping_amount,
```

tion Output

	Time	Action	Message
1	21:28:13	CREATE TABLE transactions(id VARCHAR(250), card_id VARCHAR (20), business_id VARCHAR(250), timesta...	0 row(s) affected
2	21:29:40	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__transactions.csv' INTO TA...	100000 row(s) affected Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0

Creem la taula products

```sql
CREATE TABLE products(
id VARCHAR(200) PRIMARY KEY,
product_name VARCHAR (50),
price VARCHAR(10),
colour VARCHAR(50),
weight VARCHAR(50),
warehouse_id VARCHAR (50),
category  VARCHAR(50),
brand VARCHAR(50),
cost VARCHAR(50),
launch_date VARCHAR(50));
```

ion Output

	Time	Action	Message
1	21:32:26	CREATE TABLE products(id VARCHAR(200) PRIMARY KEY, product_name VARCHAR (50), price VARCHAR(...	0 row(s) affected



Carregem dades

```
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__products.csv'
INTO TABLE products
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES;
```

Action Output

#	Time	Action	Message
1	21:32:26	CREATE TABLE products(id VARCHAR(200) PRIMARY KEY, product_name VARCHAR (50), price VARCHAR(...	0 row(s) affected
2	21:33:43	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__products.csv' INTO TABL...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0

Creen companies

```
CREATE TABLE companies(
company_id VARCHAR (250) PRIMARY KEY,
company_name VARCHAR (50),
phone VARCHAR (50),
email VARCHAR (50),
country VARCHAR (50),
website VARCHAR (255),
merchant_category VARCHAR (50),
merchant_price_position VARCHAR (50));
```

Action Output

	Time	Action	Message
1	21:32:26	CREATE TABLE products(id VARCHAR(200) PRIMARY KEY, product_name VARCHAR (50), price VARCHAR(...	0 row(s) affected
2	21:33:43	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__products.csv' INTO TABL...	100 row(s) affected Records: 100 Deleted: 0 Skipped
3	21:34:41	CREATE TABLE companies(company_id VARCHAR (250) PRIMARY KEY, company_name VARCHAR (50), ph...	0 row(s) affected

Carregem dades

```
LOAD DATA INFILE"C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__companies.csv"
INTO TABLE companies
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES;
```

Action Output

#	Time	Action	Message
1	21:36:23	LOAD DATA INFILE"C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__companies.csv" INTO TA...	100 row(s) affected Records: 100 Deleted: 0 Skipped: 0 Warnings: 0

Creen la taula europe_users

```sql
CREATE TABLE europe_users(
id VARCHAR (255) PRIMARY KEY,
name VARCHAR(50),
surname VARCHAR(50),
phone VARCHAR(50),
email VARCHAR(50),
birth_date VARCHAR(50),
country VARCHAR(50),
city VARCHAR(50),
postal_code VARCHAR(50),
address VARCHAR(50),
signup_date VARCHAR(50),
user_segment VARCHAR(50),
income_band VARCHAR(50));
LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv"
INTO TABLE europe_users
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES;
```

Output

Action Output

#	Time	Action	Message
1	21:39:58	CREATE TABLE europe_users(id VARCHAR (255) PRIMARY KEY, name VARCHAR(50), surname VARCHAR(5...	0 row(s) affected

Carreguem dades

```sql
LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv"
INTO TABLE europe_users
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES;
```

Action Output

#	Time	Action	Message
1	21:39:58	CREATE TABLE europe_users(id VARCHAR (255) PRIMARY KEY, name VARCHAR(50), surname VARCHAR(5...	0 row(s) affected
2	21:41:34	LOAD DATA INFILE "C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__european_users.csv" INT...	3990 row(s) affected Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0

Creeem america_users

```
89
90 • CREATE TABLE american_users(
91     id VARCHAR (255),
92     name VARCHAR (255),
93     surname VARCHAR (50),
94     phone VARCHAR(50),
95     email VARCHAR (50),
96     birth_date VARCHAR(50),
97     country VARCHAR (50),
98     city VARCHAR(50),
99     postal_code VARCHAR(50),
00     address VARCHAR (100),
01     signup_date VARCHAR(50),
02     user_segment VARCHAR (50),
03     income_band VARCHAR(50));
04
```

utput

Action Output

#	Time	Action	Message
1	21:47:19	CREATE TABLE american_users(id VARCHAR (255), name VARCHAR (255), surname VARCHAR (50), phone V...	Error Code: 1824. Failed to open the referencec
2	21:47:38	CREATE TABLE american_users(id VARCHAR (255), name VARCHAR (255), surname VARCHAR (50), phone V...	0 row(s) affected

Afegim dues columnas a les taules de europe_user i american_users per que las juntarem pero no volem perdre la informacio del contient

```
12 •  ALTER TABLE europe_users
13     ADD COLUMN continent VARCHAR(50);
14 •  ALTER TABLE american_users
```

tput

Action Output

#	Time	Action	Message
1	21:50:24	ALTER TABLE europe_users ADD COLUMN continent VARCHAR(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
2	21:50:28	ALTER TABLE america_users ADD COLUMN continent VARCHAR(50)	Error Code: 1146. Table 'exercici8.america_users' doesn't exist
3	21:50:41	ALTER TABLE american_users ADD COLUMN continent VARCHAR(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

Per afegir les columnas europe i america desconectem la seguretat rapidament i breument SET SQL_SAFE_UPDATES = 0; seguit de SET SQL_SAFE_UPDATES = 1;

```
UPDATE europe_users
SET continent = 'europe';
UPDATE american_users
SET continent = 'america';
```

ion Output

	Time	Action	Message
6	21:54:16	UPDATE europe_users SET continent = 'europe'	3990 row(s) affected Rows matched: 3990 Changed: 3990 Warnings: 0
7	21:54:21	UPDATE american_users SET continent = 'america'	1010 row(s) affected Rows matched: 1010 Changed: 1010 Warnings: 0
8	21:55:43	SET SQL_SAFE_UPDATES = 1	0 row(s) affected

Creen una nova taula per facilitar la tasca fusionant les dues

```sql
CREATE TABLE total_users(
id VARCHAR (255) PRIMARY KEY,
name VARCHAR (255),
surname VARCHAR (50),
phone VARCHAR(50),
email VARCHAR (50),
birth_date VARCHAR(50),
country VARCHAR (50),
city VARCHAR(50),
postal_code VARCHAR(50),
address VARCHAR (100),
signup_date VARCHAR(50),
user_segment VARCHAR (50),
income_band VARCHAR(50),
continent VARCHAR(50));
```

Action Output

#	Time	Action	Message
1	22:00:55	CREATE TABLE total_users(id VARCHAR (255) PRIMARY KEY, name VARCHAR (255), surname VARCHAR (5...	0 row(s) affected

---

Carreguem las dos taules en una

```sql
INSERT INTO total_users
SELECT * FROM europe_users
UNION ALL
SELECT * FROM american_users;
```

Action Output

#	Time	Action	Message
1	22:00:55	CREATE TABLE total_users(id VARCHAR (255) PRIMARY KEY, name VARCHAR (255), surname VARCHAR (5...	0 row(s) affected
2	22:01:26	INSERT INTO total_users SELECT * FROM europe_users UNION ALL SELECT * FROM american_users	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0

---

esborrem les que no nesesitem

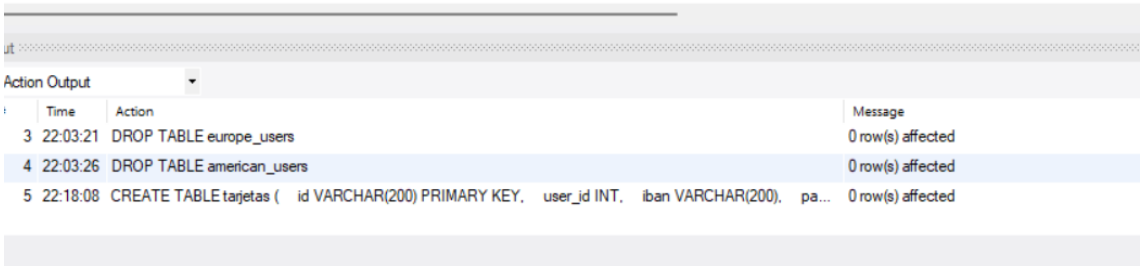
```sql
DROP TABLE europe_users;
DROP TABLE american_users;
```

Action Output

#	Time	Action	Message
2	22:01:26	INSERT INTO total_users SELECT * FROM europe_users UNION ALL SELECT * FROM american_users	5000 row(s) affected Records: 5(
3	22:03:21	DROP TABLE europe_users	0 row(s) affected
4	22:03:26	DROP TABLE american_users	0 row(s) affected

Creem la taula de tarjetas

```sql
CREATE TABLE tarjetas (
    id VARCHAR(200) PRIMARY KEY,
    user_id INT,
    iban VARCHAR(200),
    pan VARCHAR(200),
    pin VARCHAR(200),
    cvv VARCHAR(200),
    track1 VARCHAR(200),
    track2 VARCHAR(200),
    expiring_date VARCHAR(200),
    card_type VARCHAR(200),
    card_renewal_flag INT
);
```

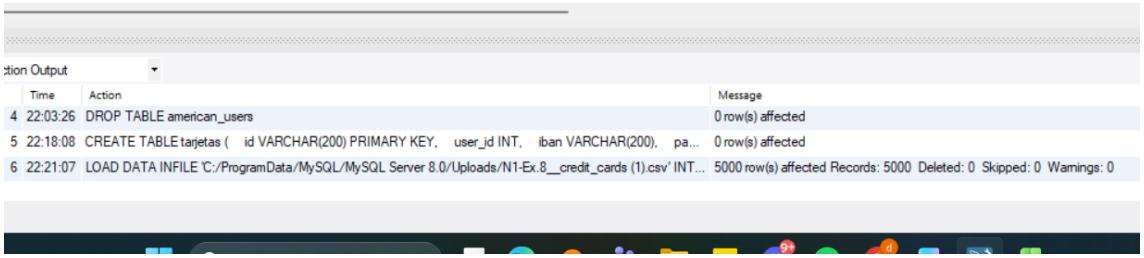


Action Output

	Time	Action	Message
3	22:03:21	DROP TABLE europe_users	0 row(s) affected
4	22:03:26	DROP TABLE american_users	0 row(s) affected
5	22:18:08	CREATE TABLE tarjetas (id VARCHAR(200) PRIMARY KEY, user_id INT, iban VARCHAR(200), pa...	0 row(s) affected

Carreguem dades

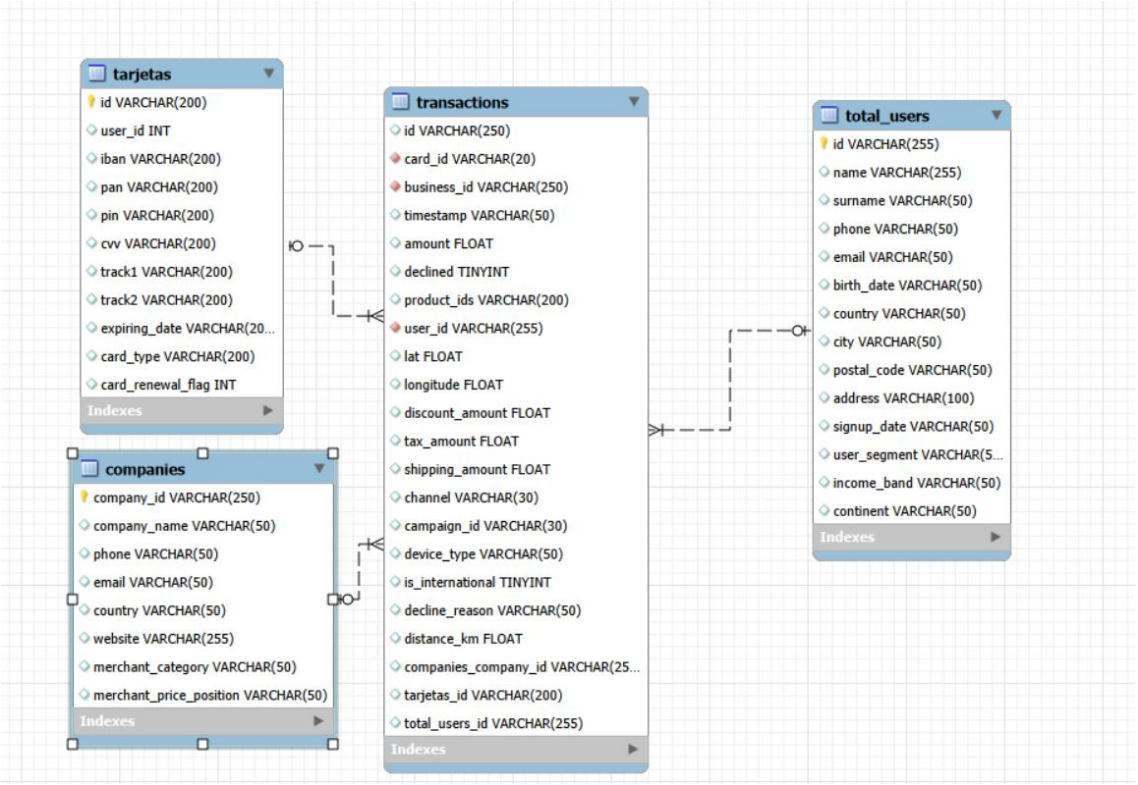
```sql
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__credit_cards (1).csv'
INTO TABLE tarjetas
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
```



tion Output

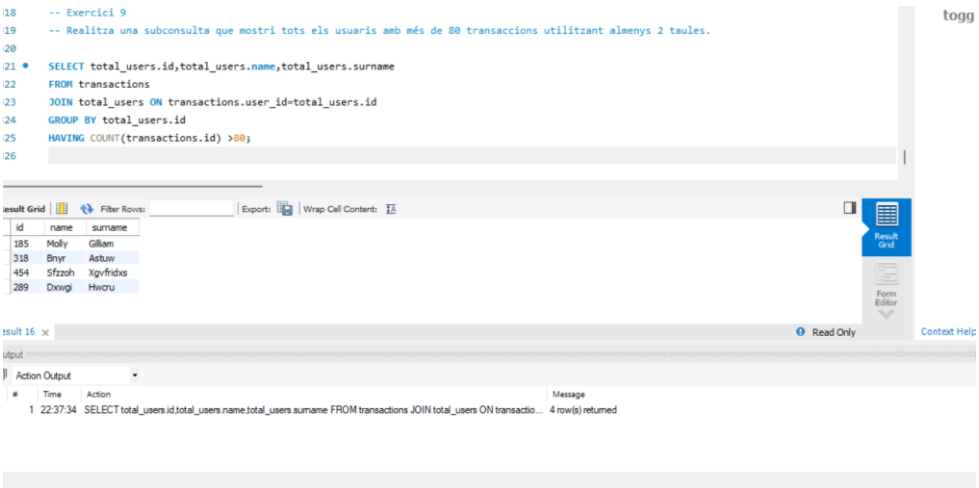
	Time	Action	Message
4	22:03:26	DROP TABLE american_users	0 row(s) affected
5	22:18:08	CREATE TABLE tarjetas (id VARCHAR(200) PRIMARY KEY, user_id INT, iban VARCHAR(200), pa...	0 row(s) affected
6	22:21:07	LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/N1-Ex.8__credit_cards (1).csv' INT...	5000 row(s) affected Records: 5000 Deleted: 0 Skipped: 0 Warnings: 0

Unim totes les taules exepte la de products



Exercici 9

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.



Exercici 10

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```sql
-- Exercici 10
-- Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2
SELECT transactions.card_id, ROUND(AVG(transactions.amount),2) as media_amount
FROM transactions
JOIN companies ON transactions.business_id=companies.company_id
WHERE companies.company_name="Donec Ltd" and
      transactions.declined=0
GROUP BY transactions.card_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

card_id	media_amount
CcU-4681	144.4
CcS-9421	131.29
CcS-6878	217.26
CcS-8677	534.55
CcS-5405	326.52
CcS-8085	407.17

Refresh data re-executing the original query

Result 18

Output

Action Output

#	Time	Action	Message
1	22:41:00	SELECT transactions.card_id, ROUND(AVG(transactions.amount),2) as media_amount FROM transactions JOI...	367 row(s) returned

NIVELL 2

EXERCICI 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

Separem de la columna timestamp día i hora

```sql
SELECT
    timestamp,
    DATE(timestamp) AS dia,
    TIME(timestamp) AS hora
FROM transactions;
```

t Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

mestamp	dia	hora
)18-12-12 08:05:17	2018-12-12	08:05:17.000000
)24-05-20 22:49:08	2024-05-20	22:49:08.000000
)18-11-04 23:16:18	2018-11-04	23:16:18.000000
)22-06-17 09:17:25	2022-06-17	09:17:25.000000
)20-11-29 07:29:16	2020-11-29	07:29:16.000000
)21-05-24 22:28:26	2021-05-24	22:28:26.000000

:20

ction Output

Time	Action	Message
1 22:58:18	SELECT timestamp, DATE(timestamp) AS dia, TIME(timestamp) AS hora FROM transactions	100000 row(s) returned

Afegim les columnas a data

```sql
ALTER TABLE transactions
ADD COLUMN dia DATE,
ADD COLUMN hora TIME;
```

Output

Time	Action	Message
23:00:39	ALTER TABLE transactions ADD COLUMN dia DATE, ADD COLUMN hora TIME	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

Carregem les dades desactivan i activa la seguretat al mateix moment

```sql
UPDATE transactions
SET
dia = DATE(timestamp),
hora = TIME(timestamp);


select*  from transactions;
SELECT dia, SUM(amount)
```

Action Output

#	Time	Action	Message
12	23:11:17	SET SQL_SAFE_UPDATES = 0	0 row(s) affected
13	23:11:21	UPDATE transactions SET dia = DATE(timestamp), hora = TIME(timestamp)	100000 row(s) affected Rows matched: 100000 Changed: 100000 Warnings: 0
14	23:11:32	SET SQL_SAFE_UPDATES = 1	0 row(s) affected

Tenim el resultat

```sql
SELECT dia, ROUND(SUM(amount),2)
FROM transactions
GROUP BY dia
ORDER BY SUM(amount) DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dia	ROUND(SUM(amount),2)
2019-11-18	15840.24
2023-02-20	15728.4
2022-12-13	15129.2
2019-03-18	14896.47
2017-12-20	14367.96

Exercici 2

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

```
SELECT companies.company_name, companies.phone, companies.country,transactions.amount,transactions.dia
FROM transactions
JOIN companies ON transactions.business_id=companies.company_id
WHERE transactions.declined=0 and
transactions.amount>350 and
transactions.amount<400 and
transactions.dia in("2015-04-29","2018-06-20","2024-03-13")
ORDER BY transactions.amount DESC;
```

Grid | Filter Rows: | Export: | Wrap Cell Content:

mpany_name	phone	country	amount	dia
quet Sem Limited	+44 744 425 9915	Netherlands	396.74	2018-06-20
ngilla LLC	+1 150 784 1716	New Zealand	390.33	2015-04-29
ctor Mauris Vel LLP	+32 133 447 8239	United States	383.58	2024-03-13
Semper Foundation	+10 283 500 6449	Sweden	371.33	2018-06-20
uris Institute	+75 473 619 9472	Sweden	369.2	2015-04-29
nc Ac PC	+15 291 730 9905	Sweden	368.63	2018-06-20

30 ×

tion Output

	Time	Action	Message
18	23:33:01	SELECT companies.company_name, companies.country,transactions.dia,transactions.amount FROM transacti...	8 row(s) returned
19	23:34:18	SELECT companies.company_name, companies.country,transactions.dia,transactions.amount FROM transacti...	8 row(s) returned
20	23:36:12	SELECT companies.company_name, companies.phone, companies.country,transactions.amount,transactions....	8 row(s) returned

Exercici 3

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.

Creen una columna per respondre a la pregunta

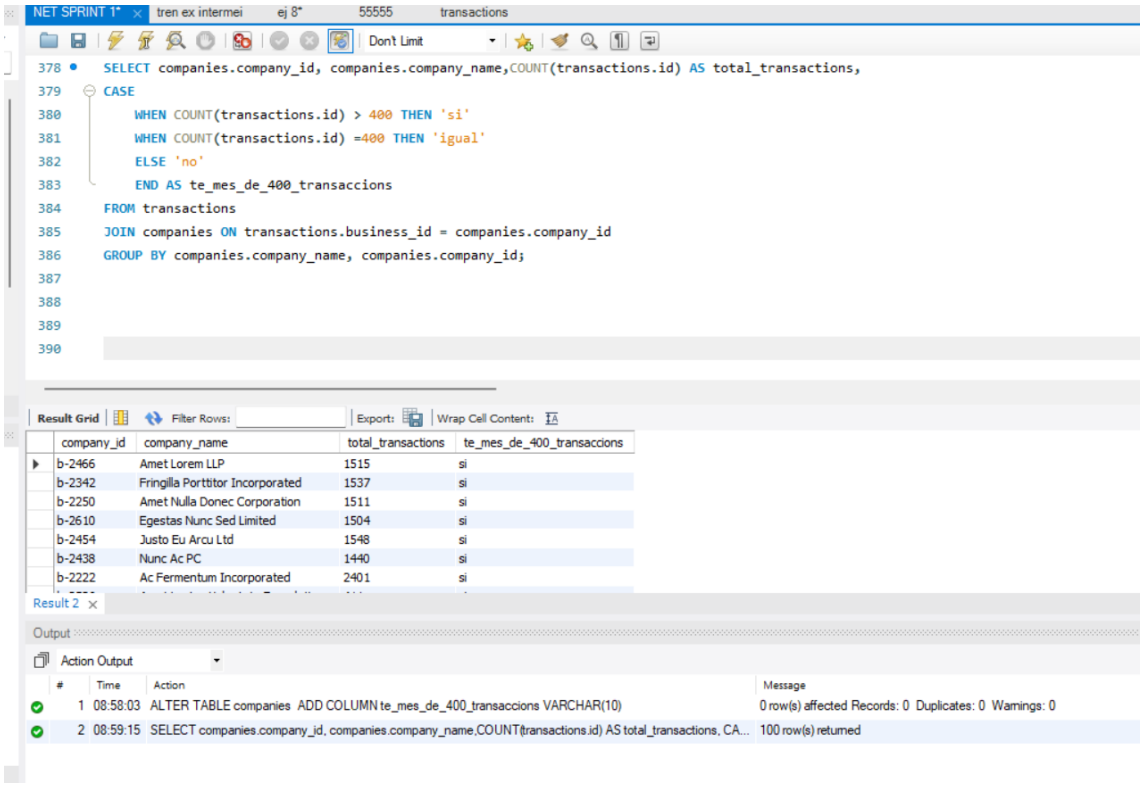
```
374
375
376 •  ALTER TABLE companies
377    ADD COLUMN te_mes_de_400_transaccions VARCHAR(10);
```

Output

Action Output

#	Time	Action	Message
1	08:58:03	ALTER TABLE companies ADD COLUMN te_mes_de_400_transaccions VARCHAR(10)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

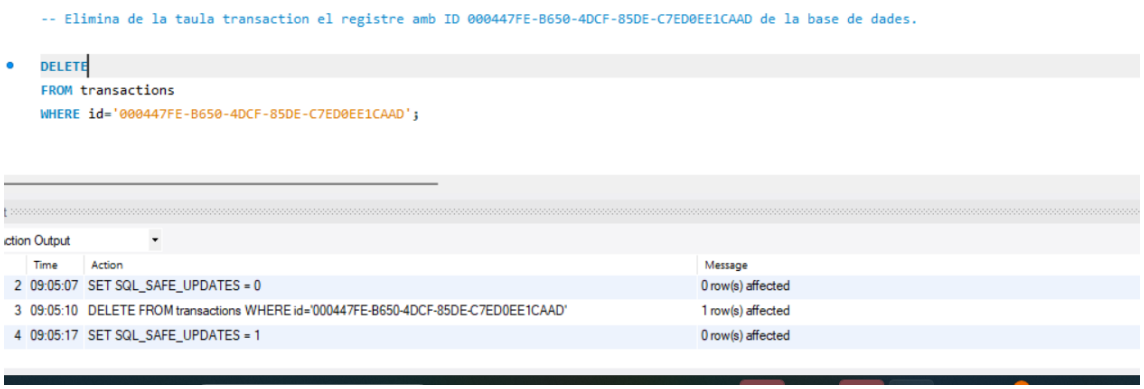
Creem un l'equivalent del IF a SQL que es el CASE WHEN(condición)THEN"="resultat



Exercici 4

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

Desactivem seguretat, borrem registre, activem seguretat i es borrem el codi deseguretat per evitar error

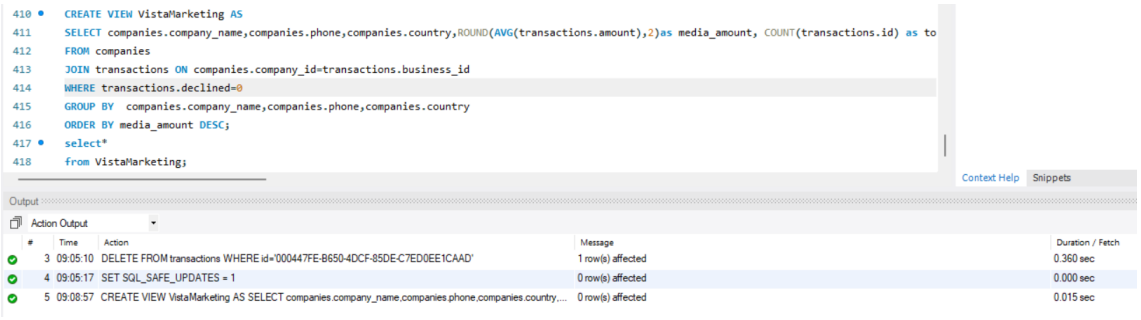


Exercici 5

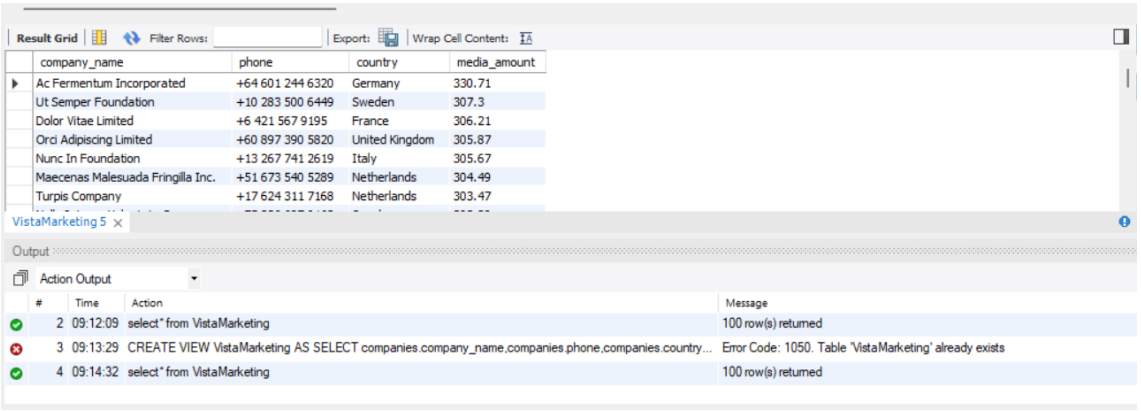
La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra

realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.

Creen la vista temporal amb CREATE VIEW i tenim en compte el declined



Cridem amb un select i veiem que es veuen les dades ordenades i requerides



d